Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 3 · SLO 2 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 2 : Session â€“ 3 : SLO â€“ 2 | Introduction to Linked Lists

I. Fill in the Blanks

1. sequence (chain/collection) â€” a linked list is a sequence of data elements.
2. nodes â€” the data elements of a linked list.
3. foundation (base/building block) â€” linked lists underlie stacks, queues and their variants.
4. chain (sequence) of nodes â€” each node holds data field(s) and a pointer to the next node.
5. self-referential structure â€” a node points to another node of the same type.
6. address of the first node â€” START stores the first node's address.
7. START â€” traversal begins from START, following next pointers.
8. list is empty â€” START = NULL means the linked list is empty.
9. data (records/elements) â€” efficient storage with insertion, deletion and updation at the cost of extra pointer space.
10. C implementation: struct node { int data; struct node *next; };